A/C EVAPORATOR (Continued)

The A/C evaporator can only be serviced by removing and disassembling the HVAC housing.

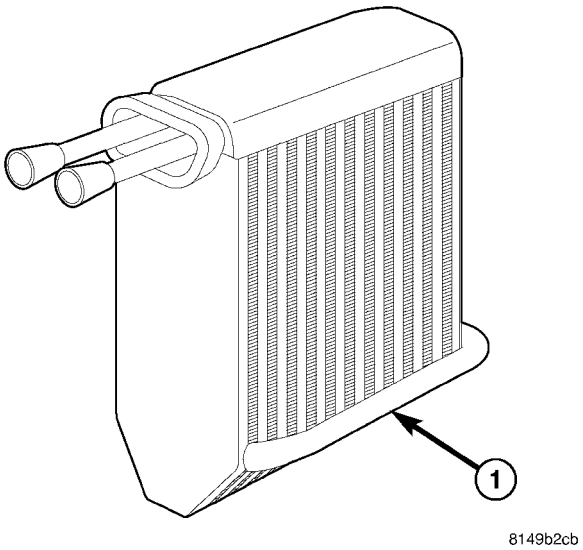

Fig. 12 A/C Evaporator - Typical

1 - A/C EVAPORATOR

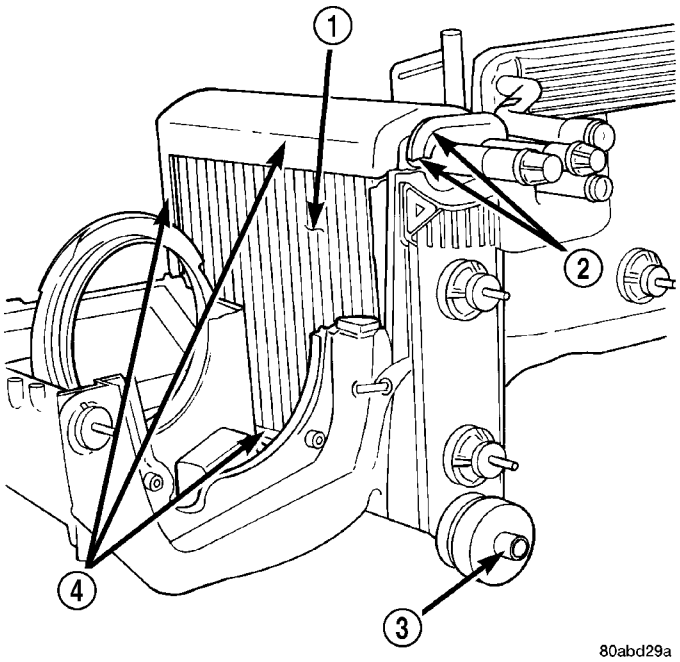

Fig. 13 A/C Evaporator - Typical

1 - A/C EVAPORATOR
2 - RUBBER GROMMET
3 - CONDENSATE DRAIN TUBE
4 - FOAM INSULATOR

OPERATION

Refrigerant enters the A/C evaporator from the A/C orifice tube as a low-temperature, low-pressure mixture of liquid and gas. As air flows over the fins of the A/C evaporator, the humidity in the air condenses on the fins, and the heat from the air is absorbed by the refrigerant. Heat absorption causes the refrigerant to boil and vaporize. The refrigerant becomes a low-pressure gas when it leaves the A/C evaporator.

The A/C evaporator cannot be repaired and, if faulty or damaged, it must be replaced.

REMOVAL

(1) Remove the HVAC housing (Refer to 24 - HEATING & AIR CONDITIONING/DISTRIBUTION/HVAC HOUSING - REMOVAL).

(2) Disassemble the HVAC housing as necessary to access the A/C evaporator (Refer to 24 - HEATING & AIR CONDITIONING/DISTRIBUTION/HVAC HOUSING - DISASSEMBLY).

(3) Carefully lift the A/C evaporator and insulator out of the lower half of the HVAC housing (Fig. 13).

(4) If required, remove the rubber grommet and the foam insulator from the A/C evaporator. If the insulator is deformed or damaged, it must be replaced.

INSTALLATION

NOTE: If the A/C evaporator is being replaced, add 45 milliliters (1.5 fluid ounces) of refrigerant oil to the refrigerant system. Use only refrigerant oil of the type recommended for the A/C compressor in the vehicle.

(1) If removed, install the foam insulator over the A/C evaporator. If the insulator is deformed or damaged, it must be replaced.

(2) Install the A/C evaporator into the lower half of the HVAC housing. Make sure that the evaporator drain within the HVAC housing is clean and unrestricted and that the foam insulator and rubber grommet are properly installed.

(3) Assemble the HVAC housing (Refer to 24 - HEATING & AIR CONDITIONING/DISTRIBUTION/HVAC HOUSING - ASSEMBLY).

(4) Install the HVAC housing (Refer to 24 - HEATING & AIR CONDITIONING/DISTRIBUTION/HVAC HOUSING - INSTALLATION).

A/C ORIFICE TUBE

DESCRIPTION

The fixed A/C orifice tube is installed in the section of the A/C liquid line closest to the A/C condenser and provides a restriction in the liquid refrigerant line between the A/C condenser and the A/C evaporator. This restriction established the pressure differential between the high and low-pressure sides of the A/C system.

The inlet end of the fixed A/C orifice tube has a nylon mesh filter screen, which filters the refrigerant